OS Theory Assignment 1

National University of Computer & Emerging Sciences, Karachi

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Task: Add a Custom 'Hello World' System Call to Linux Kernel 6.13

Objective

This document details every step taken to: (1) download Linux kernel 6.13, (2) rename it with name [Adil] so that uname -r displays the name, (3) add a custom 'Hello World' system call, (4) compile and install the kernel, and (5) verify the system call works correctly.

Step 1 – Install Required Build Dependencies

Before downloading the kernel, all tools needed to configure and compile it were installed:

```
sudo apt-get update
sudo apt-get install -y build-essential libncurses5-dev libncurses-dev bison
flex libssl-dev libelf-dev bc
```

```
system@system:~$ uname -r
6.17.0-14-generic
system@system:~$ sudo apt-get update
```

```
system@system:~$ sudo apt-get install -y build-essential libncurses5-dev libncur
ses-dev bison flex libssl-dev libelf-dev bc
Reading package lists... Done
```

Step 2 – Download the Linux Kernel Source

Kernel 6.13 was downloaded from kernel.org using wget and extracted:

```
wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.13.tar.xz
tar xf linux-6.13.tar.xz
cd linux-6.13
```

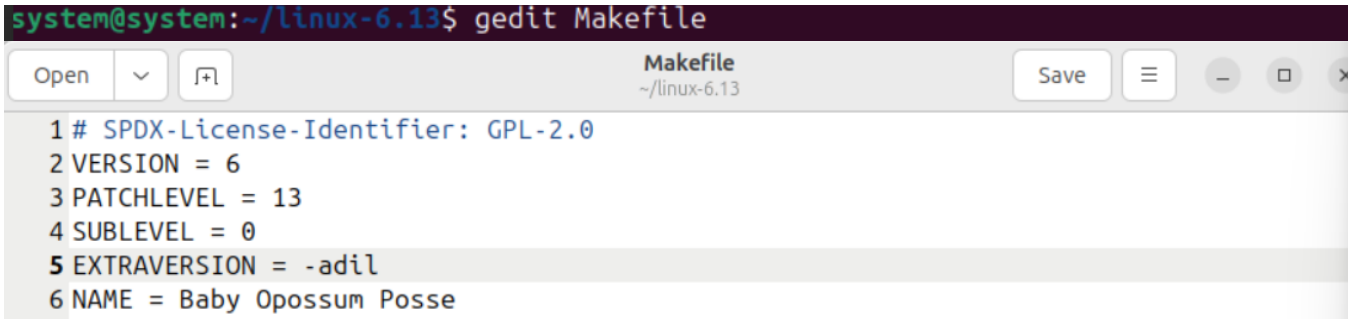
```
system@system:~$ wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.13.tar.xz
--2026-03-16 11:18:22--  https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.13.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 199.232.113.176, 2a04:4e42:21::432

system@system:~$ tar xf linux-6.13.tar.xz
system@system:~$ cd linux-6.13
```

Step 3 – Rename the Kernel with Roll Number

The EXTRAVERSION field in the top-level Makefile was edited to set the kernel version name to the roll number:

```
gedit Makefile
EXTRAVERSION = -adil
```

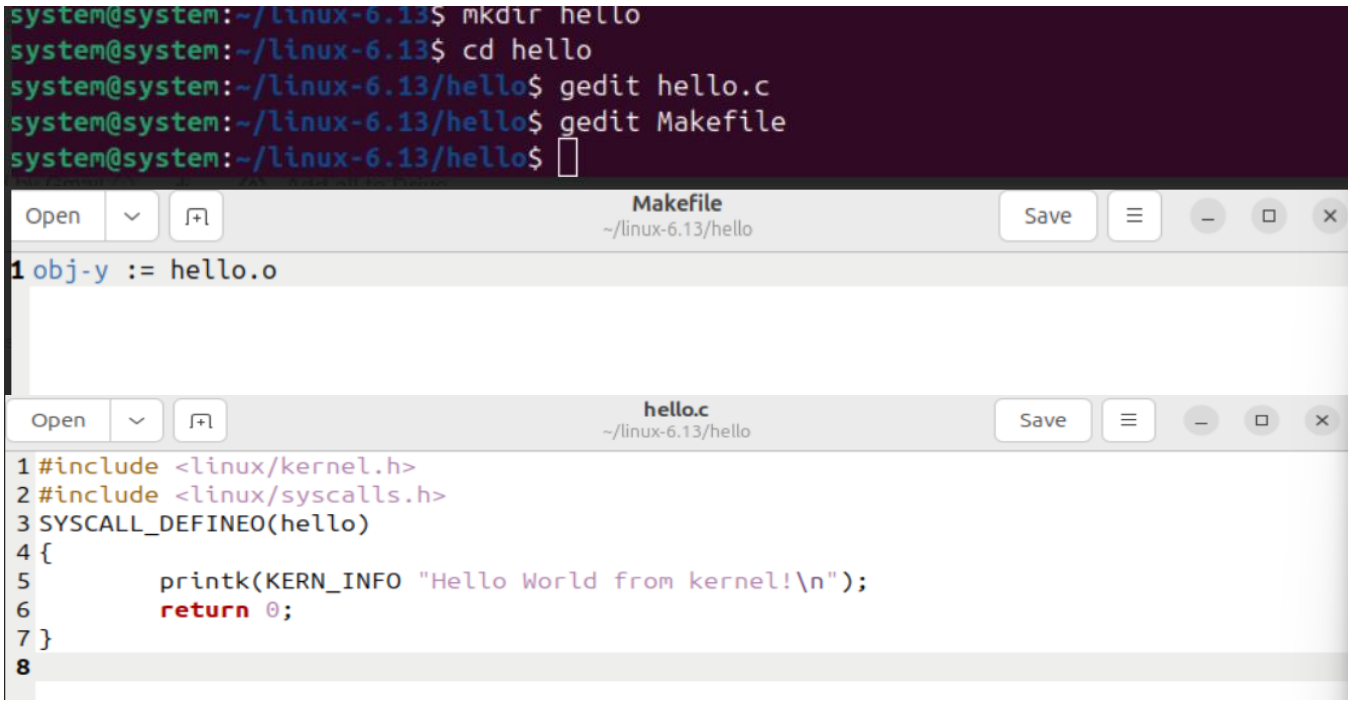


Step 4 – Add the 'Hello World' System Call

4a – Create the System Call Source File

A new directory was created and the system call implemented:

```
mkdir hello
cd hello
gedit hello.c
gedit Makefile
```



4b– Register hello/ in the Top-Level Makefile

In kernel 6.13 the core-y variable is initialised empty inside an ifeq block. It was updated to include hello/:

```
core-y      := hello/
```

```
Open  Makefile  ~/linux-6.13  Save
774 export KBUILD_MODULES KBUILD_BUILTIN
775
776 ifdef need-config
777 include $(objtree)/include/config/auto.conf
778 endif
779
780 ifeq ($(KBUILD_EXTMOD),)
781 # Objects we will link into vmlinux / subdirs we need to visit
782 core-y          := hello/
783 drivers-y       :=
784 libs-y          := lib/
785 endif # KBUILD_EXTMOD
786
```

4d – Add the System Call to the System Call Table

The syscall table at arch/x86/entry/syscalls/syscall_64.tbl was edited. The common section ended at entry 466, so the hello syscall was added as entry 467:

Gedit arch/x86/entry/syscalls/syscall_64.tbl

```
467     common  hello                   sys_hello
```

The function prototype was added to include/linux/syscalls.h just before the final #endif:

Cd include/linux

Gedit syscalls.h

```
asmlinkage long sys_hello(void);
```

```
ystem@system:~$ cd linux-6.13
ystem@system:~/linux-6.13$ gedit Makefile
ystem@system:~/linux-6.13$ gedit arch/x86/entry/syscalls/syscall_64.tbl
ystem@system:~/linux-6.13$ cd include/linux/
ystem@system:~/linux-6.13/include/linux$ gedit syscalls.h
ystem@system:~/linux-6.13/include/linux$ cd ../..
ystem@system:~/linux-6.13$ cp /boot/config-$(uname -r) .config
ystem@system:~/linux-6.13$ scripts/config --disable SYSTEM_TRUSTED_KEYS
ystem@system:~/linux-6.13$ scripts/config --disable SYSTEM_REVOCATION_KEYS
ystem@system:~/linux-6.13$ make olddefconfig
 HOSTCC  scripts/basic/fixdep
 HOSTCC  scripts/kconfig/conf.o
```

```
Open   *syscall_64.tbl   Save
~/linux-6.13/arch/x86/entry/syscalls
387 461     common  lsm_list_modules        sys_lsm_list_modules
388 462     common  mseal                   sys_mseal
389 463     common  setxattrat              sys_setxattrat
390 464     common  getxattrat              sys_getxattrat
391 465     common  listxattrat             sys_listxattrat
392 466     common  removexattrat           sys_removexattrat
393 467     common  hello                   sys_hello
394 #
395 # Due to a historical design error, certain syscalls are numbered differently
396 # in x32 as compared to native x86_64.  These syscalls have numbers 512-547.
397 # Do not add new syscalls to this range.  Numbers 548 and above are available
398 # for non-x32 use.
```

```
1306 long ksys_shmdt(char __user *shmaddr);
1307 long ksys_old_shmctl(int shmid, int cmd, struct shmid_ds __user *buf);
1308 long compat_ksys_semtimedop(int semid, struct sembuf __user *tsems,
1309                            unsigned int nsops,
1310                            const struct old_timespec32 __user *timeout);
1311 long __do_semtimedop(int semid, struct sembuf *tsems, unsigned int nsops,
1312                      const struct timespec64 *timeout,
1313                      struct ipc_namespace *ns);
1314
1315 int __sys_getsockopt(int fd, int level, int optname, char __user *optval,
1316                 int __user *optlen);
1317 int __sys_setsockopt(int fd, int level, int optname, char __user *optval,
1318                 int optlen);
1319 asmlinkage long sys_hello(void);
1320 #endif
```

Step 5 – Configure the Kernel

The running kernel's config was copied as the base, certificate keys were disabled proactively, and defaults applied for all new options:

```
cp /boot/config-$(uname -r) .config
scripts/config --disable SYSTEM_TRUSTED_KEYS
scripts/config --disable SYSTEM_REVOCATION_KEYS
make olddefconfig
```

```
ystem@system:~/linux-6.13/include/linux$ cd ../..
ystem@system:~/linux-6.13$ cp /boot/config-$(uname -r) .config
ystem@system:~/linux-6.13$ scripts/config --disable SYSTEM_TRUSTED_KEYS
ystem@system:~/linux-6.13$ scripts/config --disable SYSTEM_REVOCATION_KEYS
ystem@system:~/linux-6.13$ make olddefconfig
 HOSTCC  scripts/basic/fixdep
 HOSTCC  scripts/kconfig/conf.o
```

```
.config:811:warning: symbol value 'm' invalid for HAVE_KVM_IRQ_BYPASS
.config:10176:warning: symbol value 'm' invalid for CROS_EC_PROTO
.config:11452:warning: symbol value 'm' invalid for ANDROID_BINDER_IPC
.config:11453:warning: symbol value 'm' invalid for ANDROID_BINDERFS
#
# configuration written to .config
#
system@system:~/linux-6.13$ scripts/config-disable SYSTEM_TRUSTED_KEYS
bash: scripts/config-disable: No such file or directory
system@system:~/linux-6.13$ scripts/config --disable SYSTEM_TRUSTED_KEYS
system@system:~/linux-6.13$ scripts/config --disable SYSTEM_REVOCATION_KEYS
system@system:~/linux-6.13$ make j$ (nproc) 2>&1 | tee build.log
```

Step 6 – Compile the Kernel

The kernel was compiled using all available CPU cores. This took approximately 2.5–3 hours on the ThinkPad T440:

```
make -j$(nproc) 2>&1 | tee build.log
```

```
system@system:~/linux-6.13$ make -j$(nproc) 2>&1 | tee build.log
  SYNC    include/config/auto.conf.cmd
*
* Restart config...
*
*
* Certificates for signature checking
*
File name or PKCS#11 URI of module signing key (MODULE_SIG_KEY) [certs/signing_key.pem] certs/si
ning_key.pem
Type of module signing key to be generated
> 1. RSA (MODULE_SIG_KEY_TYPE_RSA)
```

```
@system:~/linux-6.13$ make -j$(nproc) 2>&1 | tee build.log
    include/config/auto.conf.cmd


art config...
system@system:~/linux-6.13$ make -j$(nproc) 2>&1 | tee build.log
mkdir -p /home/system/linux-6.13/tools/objtool && make O=/home/system/linux-6.13 subdir=tools/objtool --no-p
rint-directory -C objtool
  INSTALL libsubcmd_headers
  CALL    scripts/checksyscalls.sh
  CHK     kernel/kheaders_data.tar.xz
Kernel: arch/x86/boot/bzImage is ready  (#1)
```

Step 7 – Install Modules and the Kernel

Kernel modules and the kernel image were installed into the system:

```
sudo make modules_install
sudo make install
```

```
system@system:~$ cd linux-6.13
system@system:~/linux-6.13$ sudo make modules_install
[sudo] password for system:
  SYMLINK /lib/modules/6.13.0-adil/build
  INSTALL /lib/modules/6.13.0-adil/modules.order
  INSTALL /lib/modules/6.13.0-adil/modules.builtin
```

Fig 7.2 – sudo make install output

Step 8 – Create the Initial RAM Filesystem

An initramfs image was generated for the new kernel as required by the lab manual:

```
sudo update-initramfs -c -k 6.13.0-adil
```

```
system@system:~/linux-6.13$ sudo update-initramfs -c -k 6.13.0-adil
update-initramfs: Generating /boot/initrd.img-6.13.0-adil
```

Step 9 – Update GRUB and Reboot

GRUB was updated to detect and include the new kernel, then the system was rebooted:

```
sudo update-grub
sudo reboot
```

```
system@system:~/linux-6.13$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.17.0-19-generic
Found initrd image: /boot/initrd.img-6.17.0-19-generic
Found linux image: /boot/vmlinuz-6.17.0-14-generic
Found initrd image: /boot/initrd.img-6.17.0-14-generic
```

Step 10 – Verify the New Kernel

After rebooting into the new kernel, the following verifications were performed as required by the lab manual:

Verify uname -r shows roll number

```
uname -r
```

```
system@system:~$ uname -r
6.13.0-adil
system@system:~$ ls /boot | grep 6.13
config-6.13.0-adil
initrd.img-6.13.0-adil
System.map-6.13.0-adil
vmlinuz-6.13.0-adil
system@system:~$ gedit test_hello.c
system@system:~$ gcc test_hello.c
```

Verify /boot contains new kernel files

```
ls /boot | grep 6.13.0-adil
```

```
system@system:~$ ls /boot | grep 6.13
config-6.13.0-adil
initrd.img-6.13.0-adil
System.map-6.13.0-adil
vmlinuz-6.13.0-adil
```

Step 11 – Write and Run the Test Program

```
system@system:~$ gedit test_hello.c

system@system:~$ gcc test_hello.c

system@system:~$ ls
a.out    Documents  linux-6.13         Music     Public  Templates     Videos
Desktop  Downloads  linux-6.13.tar.xz  Pictures  snap    test_hello.c
system@system:~$ ./a/out
bash: ./a/out: No such file or directory
system@system:~$ ./a.out
System call returned: 0
system@system:~$ sudo dmesg | grep Hello
[sudo] password for system:
[  386.747544] Hello World from kernel!
system@system:~$
```